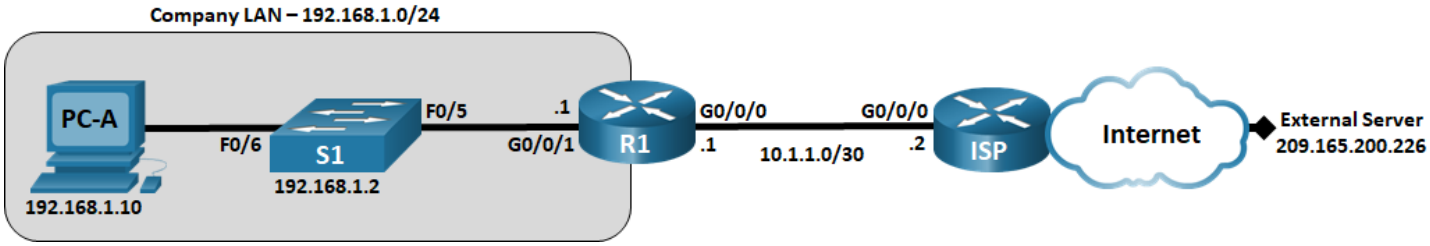
Laborationer (modul 7)
Lab - Troubleshoot Connectivity Issues

Topology



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0/1	192.168.1.1	255.255.255.0	N/A
	G0/0/0	10.1.1.1	255.255.255.252	N/A
ISP	G0/0/0	10.1.1.2	255.255.255.252	N/A
	Lo0	209.165.200.226	255.255.255.255	N/A
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.10	255.255.255.0	192.168.1.1

Objectives

Part 1: Identify the Problem

Part 2: Implement Network Changes

Part 3: Verify Full Functionality

Part 4: Document Findings and Configuration Changes

Background / Scenario

In this lab, the company that you work for is experiencing problems with their Local Area Network (LAN). You have been asked to troubleshoot and resolve the network issues. In Part 1, you will connect to devices on the LAN and use troubleshooting tools to identify the network issues, establish a theory of probable cause, and test that theory. In Part 2, you will establish a plan of action to resolve and implement a solution. In Part 3, you will verify full functionality has been restored. Part 4 provides space for you to document your troubleshooting findings along with the configuration changes that you made to the LAN devices.

Note: The routers used with CCNA hands-on labs are Cisco 4221 with Cisco IOS XE Release 16.9.4 (universalk9 image). The switches used in the labs are Cisco Catalyst 2960s with Cisco IOS Release 15.2(2) (lanbasek9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs. Refer to the Router Interface Summary Table at the end of the lab for the correct interface identifiers.

© 2016 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

Required Resources

- 2 Routers (Cisco 4221 with Cisco IOS XE Release 16.9.4 universal image or comparable)
- 1 Switch (Cisco 2960 with Cisco IOS Release 15.2(2) lanbasek9 image or comparable)
- 1 PC (Windows with a terminal emulation program, such as Tera Term)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology\

Troubleshooting Configuration

The following settings must be configured on the devices shown in the topology. Paste the configurations onto the specified devices prior to starting the lab.

S1:

```
no ip domain-lookup
hostname S1
ip domain-name ccna-lab.com
username admin01 privilege 15 secret cisco12345
interface FastEthernet0/1
 shutdown
interface FastEthernet0/2
 shutdown
interface FastEthernet0/3
 shutdown
interface FastEthernet0/4
 shutdown
interface FastEthernet0/5
 speed 10
 duplex half
interface Vlan1
 ip address 192.168.1.2 255.255.255.0
ip default-gateway 192.168.1.0
banner motd $ Authorized Users Only! $
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
crypto key generate rsa general-keys modulus 1024
end
```

R1:

```
hostname R1
no ip domain lookup
ip domain name ccna-lab.com
username admin01 privilege 15 secret cisco12345
```

© 2016 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
interface GigabitEthernet0/0/1
 ip address 192.168.1.1 255.255.255.0
 no negotiation auto
 speed 100
 no shutdown
interface GigabitEthernet0/0/0
 ip address 10.1.1.1 255.255.255.252
 no shutdown
banner motd $ Authorized Users Only! $
line vty 0 4
 login local
 transport input ssh
crypto key generate rsa general-keys modulus 1024
end
```

ISP:

```
hostname ISP
no ip domain lookup
interface GigabitEthernet0/0/0
 ip address 10.1.1.2 255.255.255.252
 no shut
interface Lo0
 ip address 209.165.200.226 255.255.255.255
ip route 0.0.0.0 0.0.0.0 10.1.1.1
end
```

Part 1: Identify the Problem.

The only available information about the network problem is that the users are experiencing slow response times and that they are not able to reach an external device on the internet at IP address 209.165.200.226. To determine probable cause(s) for these network issues, you will need to utilize network commands and tools on the LAN equipment shown in the topology.

Note: The user name **admin01** with a password of **cisco12345** will be required to log into the network equipment.

Step 1: Troubleshoot the network.

Use the tools available to you to troubleshoot the network, keeping in mind that the requirement is to restore connectivity to the external server and the eliminate slow response times.

Note: While using SSH to connect to network devices, issue the **terminal monitor** privileged exec command to enable log output to the SSH console.

Step 2: Document the probable causes.

List the probable causes for the network problems that employees are experiencing.

© 2016 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

Part 2: Implement Network Changes

You have communicated the problems that you discovered in Part 1 to your supervisor. She has approved these changes and has requested that you implement them.

Part 3: Verify Full Functionality

Verify that full functionality has been restored. PC-A, S1, and R1 should be able to reach the external server, and ping replies from PC-A to the external server should exhibit no significant variation in response times.

Part 4: Document Findings and Configuration Changes

Use the space provided below to document the issues found during your troubleshooting and the configurations changes made to resolve those issues.here.

Reflection Question

This lab had you troubleshoot all devices before making any changes. Is there another way to apply the troubleshooting methodology?

Glöm inte att rensa enheterna när du är färdig!

© 2016 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public